

# PRJ301 Demo Learning Material

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## 1. Java Server Pages (JSP)

JSP is a server-side view technology for Java web applications. A JSP page combines static HTML with JSP tags and Expression Language. The web container translates the JSP file into a servlet class, compiles it, creates the servlet instance, and executes it to generate an HTTP response.

### JSP request lifecycle

- Translation: the container converts the JSP source into Java servlet source code.
- Compilation: the generated servlet source is compiled into bytecode.
- Initialization: `jspInit()` runs once when the JSP servlet is initialized.
- Request processing: `_jspService()` runs for every request and writes the response.
- Destruction: `jspDestroy()` runs before the servlet is removed.

## 2. Spring Boot

Spring Boot helps developers create Spring applications quickly through auto-configuration, starter dependencies, embedded servers, and production-ready conventions. A typical REST application uses `@RestController` for HTTP endpoints, `@Service` for business rules, and repository components for persistence.

- Auto-configuration selects beans based on dependencies and configuration.
- Starter dependencies provide compatible libraries for common use cases.
- Embedded Tomcat allows the application to run as a standalone JAR.

# PRJ301 Review Notes

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## 3. Spring MVC request flow

A request first reaches DispatcherServlet. It selects a controller through handler mappings. The controller validates input and calls a service. The service applies business logic and may call a repository. The controller then returns a view or a serialized response body.

## 4. Layered architecture

- Controller: HTTP input, validation boundary, and response status.
- Service: business rules and transaction-oriented operations.
- Repository: database access and persistence queries.
- DTO: stable request and response contracts between frontend and backend.

## 5. Demo questions

- What happens the first time a JSP page is requested?
- Why does Spring Boot use starter dependencies?
- What is the role of DispatcherServlet in Spring MVC?

## Expected learning outcome

Students should be able to explain the JSP lifecycle, identify the responsibilities of controller-service-repository layers, and describe the Spring MVC request flow without exposing personal or confidential information to external AI providers.